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PALEOMAGNETISM OF SOME PRECAMBRIAN BASALTIC FLOWS AND RED BEDS, EASTERN GRAND CANYON, ARIZONA

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Lava flows and red sandstone beds near the middle of the Upper Precambrian Grand Canyon Series exhibit stable remanent magnetization. The beds are about 1000 m stratigraphically above rocks of the Grand Canyon Series for which paleomagnetic poles have been reported. All specimens were subjected to stepwise thermal (200°–700°C) or alternating field (25–5000 Oe) demagnetization for the determination of characteristic magnetization. The pole for two flows and an intercalated sandstone bed of the Cardenas Lavas of Ford, Breed and Mitchell (upper Unkar Group), is at 174.6W, 0.4N (N = 10, K = 50, α_{95} = 6.9°). The pole for a weathered zone developed across the Cardenas Lavas is at 167.8W, 49.4N (N = 5, K = 79, α_{95} = 8.6°). The pole for directly overlying sandstone of the Nankoweap Formation of Maxson is at 174.4E, 12.5N (N = 6, K = 105, α_{95} = 6.6°). These poles lie on or near, and appear to follow, part of an apparent polar wandering path recently proposed for the Precambrian of North America by Spall. If the fit is not accidental, little or no rotation has occurred between north-central Arizona and parts of the North American continent used to define the proposed path.

1. Introduction

A thick section of gently dipping strata of Precambrian age, the Grand Canyon Group of Powell [1] and Grand Canyon Series of Walcott [2], is exposed in the eastern part of the Grand Canyon, northern Arizona. Rocks of the Grand Canyon Series total at least 3350 m in thickness and comprise intervals of sandstone, shale, limestone and lava flows that are assigned to various formations and groups (table 1). In the lower part of the Unkar Group, the Bass Limestone, Hakatai Shale, and Shinumo Quartzite have previously been sampled for paleomagnetic analysis [4–7]. We present data from samples collected during a preliminary study of basaltic andesite flows and intercalated red sandstone in the upper part of the Unkar Group (Cardenas Lavas of Ford and others, [10]), and directly overlying red sandstone of the Nankoweap Formation of Maxson [3].

Precambrian apparent polar wandering, as determined from North American data, has recently been synthesized by Spall [11], Robertson and Fahrig [12], and Gates [13]. Paleomagnetic poles from isotopically dated rocks lead Spall to identify five sequential groups of pole positions in the interval between 1000 and 2485 my ago.

An average pole position for the Grand Canyon Series (more specifically, from poles of the lower Unkar Group) has been given by Spall [11]. Spall stresses that poles of the lower Unkar Group are only suggestive because most samples have not been demagnetized, or have been only partially demagnetized.

2. Geologic setting

2.1. Location

The beds sampled crop out in the eastern part of Grand Canyon in the first side canyon west of Tanner Canyon, about 2 km north–northeast of Cardenas Butte. Access is by means of the Tanner Trail to a bench formed by the Tapeats Sandstone of Cambrian age, and then 0.3 km cross-country to the southwest.

2.2. Units sampled

The five stratigraphic units sampled are numbered from 1 to 5 in ascending stratigraphic order. Two formations (Cardenas Lavas and Nankoweap Formation) were sampled, as well as an intervening weathered zone. The development of geologic relations, and stratigraphic

Table 1
Generalized description of the Grand Canyon Series, northern Arizona

	Thickness (meters)
Chuar Group [2,3]; marine limestone and gray shale	1550–1610
Disconformity [8]	
Nankoweap Group of Van Gundy [8]	
Nankoweap Formation of Maxson [3]; thin-bedded red sandstone; hematitic beds at base (sample unit no. 5)	100+
Disconformity [8]	
Unkar Group [2,9]	
Cardenas Lavas of Ford and others [10]; locally preserved ferruginous weathered zone (sample unit no. 4), developed across basaltic andesite flows and subordinate interbedded sandstone (sample unit nos. 1, 2 and 3)	244–450
Dox Sandstone [9]; thin-bedded red sandstone	915
Shinumo Quartzite [9]; massive, cross-bedded, white and locally red sandstone	335–475
Hakatai Shale [9]; vermilion mudstone	177–253
Bass Limestone [9] and underlying Hotauta Conglomerate [9]; limestone and reddish sandstone	37–104
Thickness of Unkar Group	1700–2200

positions of the samples, are shown diagrammatically in fig. 1. Materials of the Nankoweap Formation sampled are ferruginous and they rest with about 5° angular discordance against the lower part of the Cardenas Lavas. A ferruginous weathered zone is developed on the lava series beneath a pre-Nankoweap erosional surface that truncates the entire section of Cardenas flows. The structural reconstruction (fig. 1) shows that the small angular discordance is most simply explained as the consequence of initial dip and differential compaction, presumably occurring as the stratigraphically lowest beds of the Nankoweap were incrementally deposited against an approximately 20° erosional escarpment developed across the lava series. Beds of the Cardenas were near-level during deposition of the Nankoweap, because beds in the upper part of the Nankoweap rest with apparent conformity on the upper flow of the lava series.

The basal flow and an intermediate level flow in the lava series (units 1 and 3, respectively) were sampled across 1.5 and 15 m stratigraphic intervals in their interior parts. The weathered zone (unit 4) was sampled where it is developed on the intermediate-level flow, which allowed the magnetization of weathered flow rock to be compared with the magnetization of apparently unweathered parent rock. The weathered zone, about 10 m thick, was sampled 2 to 3 m below its top. A two meter thick, red sandstone bed near the base of the Cardenas (unit 2), and two hematiterich sandstone beds, about 4 m thick, at the base of the Nankoweap Formation (unit 5) were sampled at closely spaced (6 cm to 1.2 m) intervals across stratigraphic sections.

3. Field and laboratory procedures

Six to eight cores, 2.54 cm in diameter and 3 to 5 cm long, were obtained from each of the five units sampled. A portable core drill and orientation equipment similar to that described by Doell and Cox [14] were used for sampling. Core azimuth and plunge were measured to the nearest one-half degree.

Remanent magnetization was measured with an air-driven spinner magnetometer, which operates at 155 Hz and has a background noise of approximately 2×10^{-7} emu. Individual cores were trimmed in the laboratory by coring twice more orthogonally to the axis of the core, producing near-spherical 9 cm^3 bodies. Intensities of magnetization of the specimens ranged from about 2×10^{-2} emu to 2×10^{-4} emu.

Specimens of each stratigraphic unit were subjected to both thermal and alternating field (AF) cleaning. Thermal demagnetization was done in air in five steps at 200, 350, 500, 625, and 700°C. AF cleaning was done in a solenoid in 4 to 10 steps between 25 and 5000 Oe peak field strength, using a three axis tumbler similar to the one described by Doell and Cox [15].

4. Paleomagnetic analysis

4.1. Natural remanent magnetism (NRM)

4.1.1. Directions and intensities

Plots of declination and inclination, and intensities

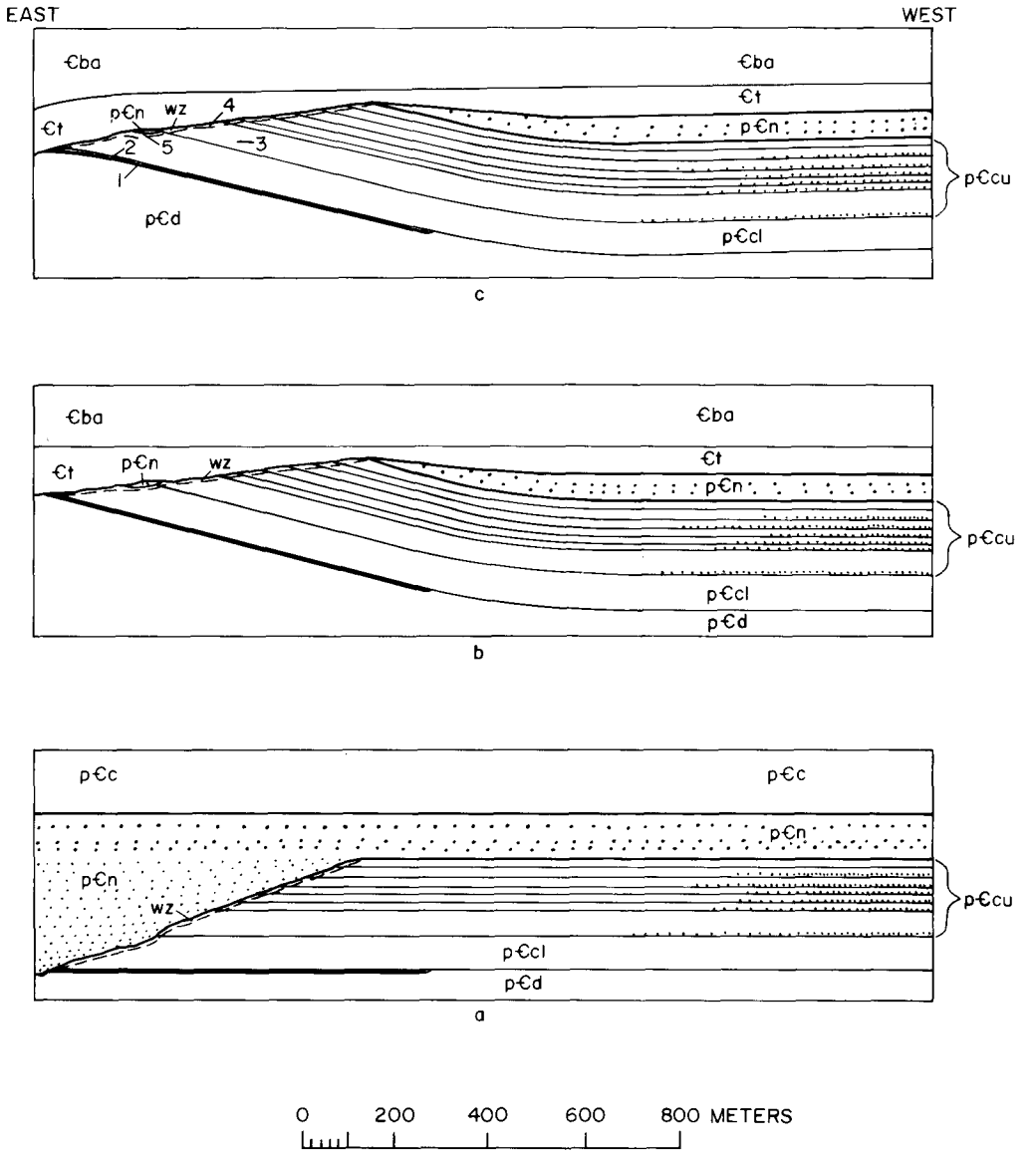


Fig. 1. Cross-section diagrams of development of structural relations in canyon west of Tanner Canyon. View is to south. No vertical exaggeration. *Cambrian rocks:* €ba, Bright Angle Shale; €t, Tapeats Sandstone. *Precambrian rocks:* p€c, Chuar Group; p€n, Nankoweap Formation (sample unit no. 5 from basal ferruginous beds); wz, ferruginous weathered zone on Cardenas Lavas (sample unit no. 4); p€cu, upper cliff-forming part of €ardenas Lavas (sample unit no. 3 from lowest flow); p€cl, lower slope-forming part of Cardenas Lavas, which includes ledge-forming lava flow at base (sample unit no. 1) and overlying Dox-like sandstone (sample unit no. 2); p€d, Dox Sandstone. (c) *Present:* Tertiary folding led to the development of the east-facing Butte or East Kaibab monocline; a small part of the upper limb is shown in the left side of the diagram. (b) *Cambrian time:* Folding after deposition of the Chuar was followed by erosion and removal of Chuar rocks and part of the Nankoweap Formation. The Cambrian Tapeats Sandstone then was deposited across the irregular Ep-Algonkian erosion surface. (a) *Chuar time:* Following deposition of the Cardenas Lavas, a prominent east-facing erosional escarpment was formed across the lava series. Following development of a ferruginous zone of weathering and cementation, a minimum of 200 m of Nankoweap is inferred to have been deposited disconformably against the scarp, before deposition of at least 100 m of sandstone on the "plateau" area to the west.

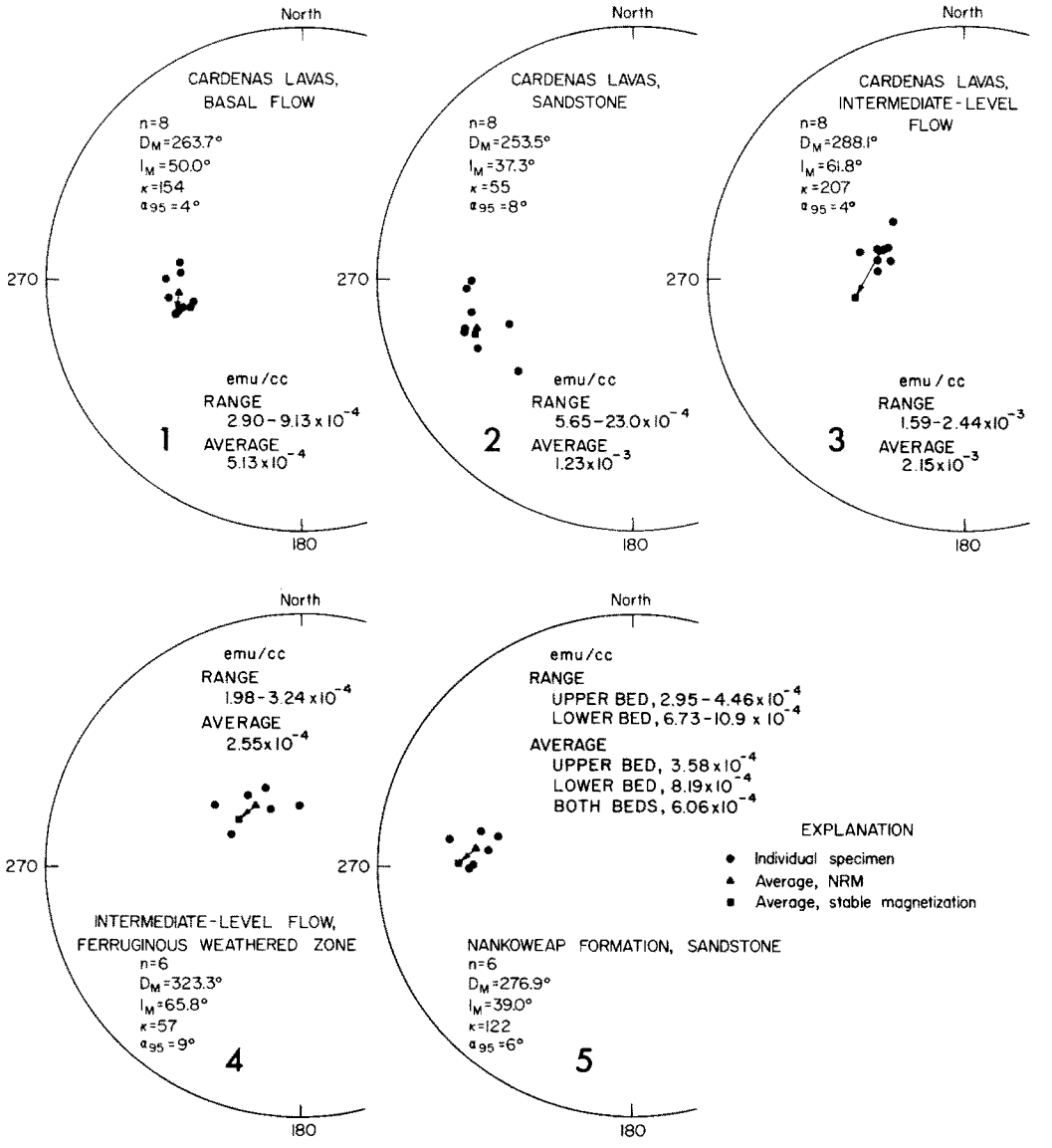


Fig. 2. Plots of declination and inclination of NRM and stable magnetization, and intensity of NRM, for Grand Canyon samples. Statistics are for the NRM data. Southerly shifts in the directions occurred with removal of the less stable components of magnetization. Small shifts occurred where the remanent magnetization resides largely in hematite (units 1, 2, 4 and 5). A large shift occurred in unit 3, an essentially unaltered flow in which magnetization resides largely in magnetite.

of NRM, are summarized in fig. 2. All specimens exhibit apparent normal polarity (north-seeking pole down and assuming that a comparatively short apparent polar wandering path connects Paleozoic and Precambrian poles). All show westward declinations. Poles for four of the units plot in the central Pacific Ocean, in positions similar to poles of lower Unkar Group formations reported by others [4,5,7]. Statistical groupings within the units are good, particularly for the basal and intermediate flows of the lava

series. In contrast, NRM plots of specimens from the two sandstone units show greater scatter, although progressive demagnetization analysis showed that the red beds are more stably magnetized than the flows. The two successive sandstone beds in the Nankoweap Formation exhibit different intensities and apparently slightly different directions of magnetization, which suggests that their remanence was acquired at the time of or shortly after deposition and is not the result of a later general re-magnetization. NRM poles of the five sampled units lie north of stable poles, along great circles that trend toward the earth's present field, which indicates that the NRM contains a component representing recent magnetization.

4.2. Demagnetization

4.2.1. Thermal demagnetization

Thermal demagnetization was carried out in air at steps of 200, 350, 500, 625, and 700°C, on two to four specimens of each unit (fig. 3). Four to seven specimens of the various units were heated together during each thermal run, which served as an internal check for recognizing the accidental imposition of a magnetic moment.

Specimens of red sandstone (units 2 and 5) were magnetically stable through the 625°C demagnetization step (fig. 3), reflecting that the magnetization resides mainly in hematite. Hematite in the lower of the two Nankoweap beds sampled occurs in depositional laminae and interstitially; hematite of the upper bed occurs interstitially.

The basal flow of the Cardenas is magnetically stable. Its field and microscopic appearance suggest substantial deuteric alteration, and secondary serpentine, hematite, chalcedony, quartz, and sericite are abundant. The 2 m thick red sandstone bed (unit 2) that lies directly on the basal flow does not appear to be altered, which suggests that alteration of the underlying flow occurred before deposition of the sandstone. Stepwise thermal demagnetization of the samples from the basal flow shows that much of the decrease in remanent magnetization takes place above 625°C, reflecting hematite with a narrow blocking temperature distribution.

Samples from the intermediate flow display the strongest natural remanent magnetization (fig. 2). The remanence resides in finely disseminated magnetite that became oxidized during the heating runs in air.

The ferruginous weathered zone displays a demagnetization curve at the 200°C and 350°C cleaning steps that is less steep than the demagnetization curve for the parent intermediate flow (fig. 3). The minerals seen in polished section (matrix altered to siderite and earthy hematite, and specular hematite replacing leucoxene) indicate that the remanence resides in the products of weathering. The remanent magnetization remaining after heating to 200° and 350°C provides usable directional information (fig. 5). The increase in intensity which occurred at the higher thermal cleaning steps (fig. 3) resulted from a weak magnetization inadvertently imposed during cooling.

4.2.2. Alternating field (AF) demagnetization

Demagnetization by alternating fields show varying degrees of stability in the five units sampled (fig. 4). The red sandstones and the basal lava flow have a very hard magnetization, whereas the magnetite-bearing intermediate flow has a comparatively soft magnetization. The weathered zone displays an intermediate hardness of magnetization, marked by a broad scatter in intensity of magnetization at the higher field demagnetization steps.

4.3. Determination of poles

Orthogonal demagnetization diagrams [16] were constructed to determine intervals of stable magnetization for use in the calculation of poles. Representative orthogonal diagrams for thermal and AF demagnetization are shown for the five units sampled (figs. 5 and 6). The two methods of cleaning provided mutually consistent directional data.

Intervals of stable magnetization are defined on the orthogonal diagrams as including all points that describe lines which trend toward the origins (zero magnetization points) of the coordinate axes. Intervals of stable magnetization determined this way were obtained for each specimen, and an average declination and inclination calculated for the defined ranges of stability. We believe such a procedure provides slightly improved results, with respect to the use of single stable demagnetization steps, because vectors for individual specimens are averaged over

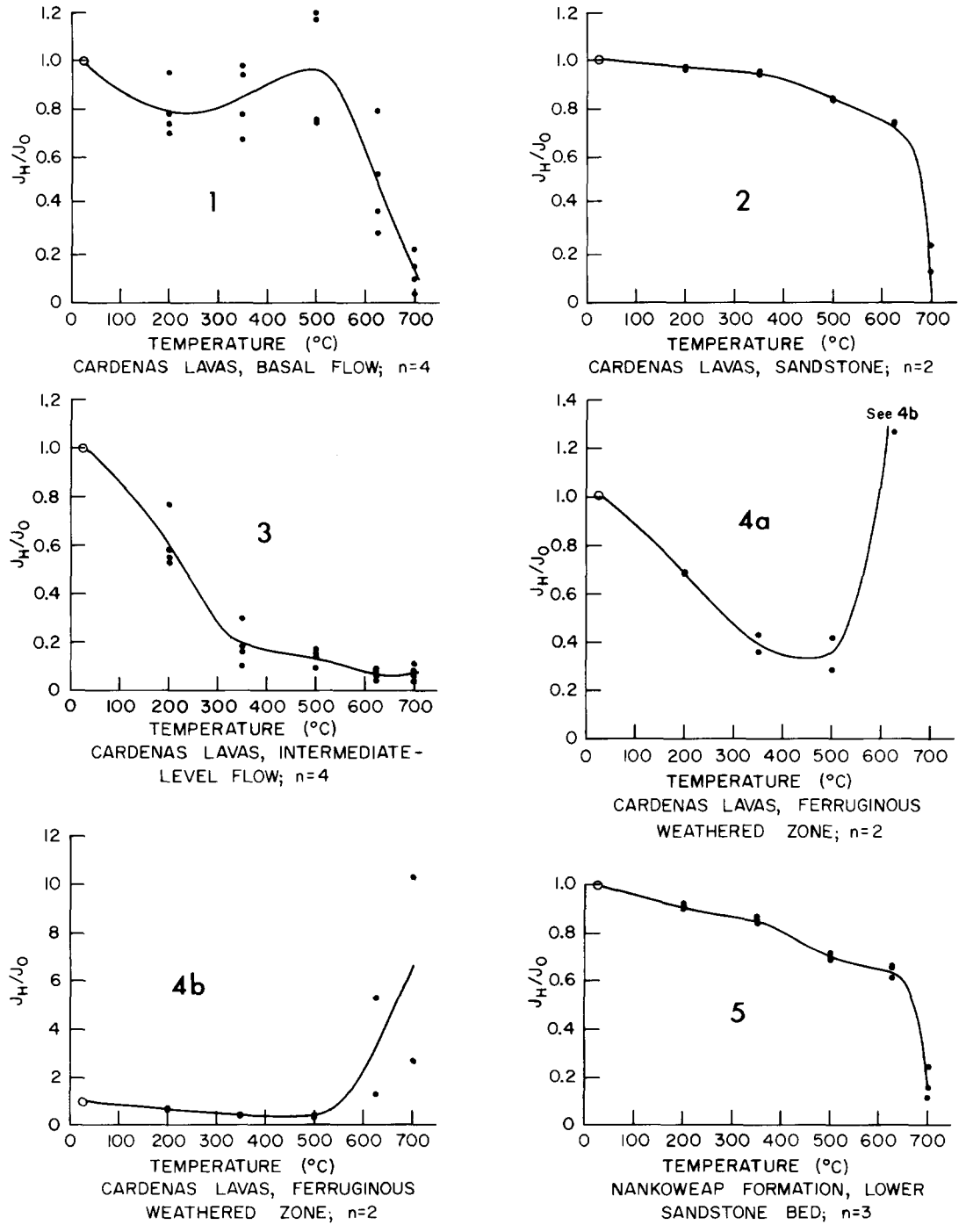


Fig. 3. Normalized thermal demagnetization curves for five Grand Canyon units sampled. Unit 4, plot b, shows data points for unit 4 plotted at a reduced verical scale. J_H/J_0: ratio of magnetization remaining after a given demagnetization step to original level of magnetization (NRM).

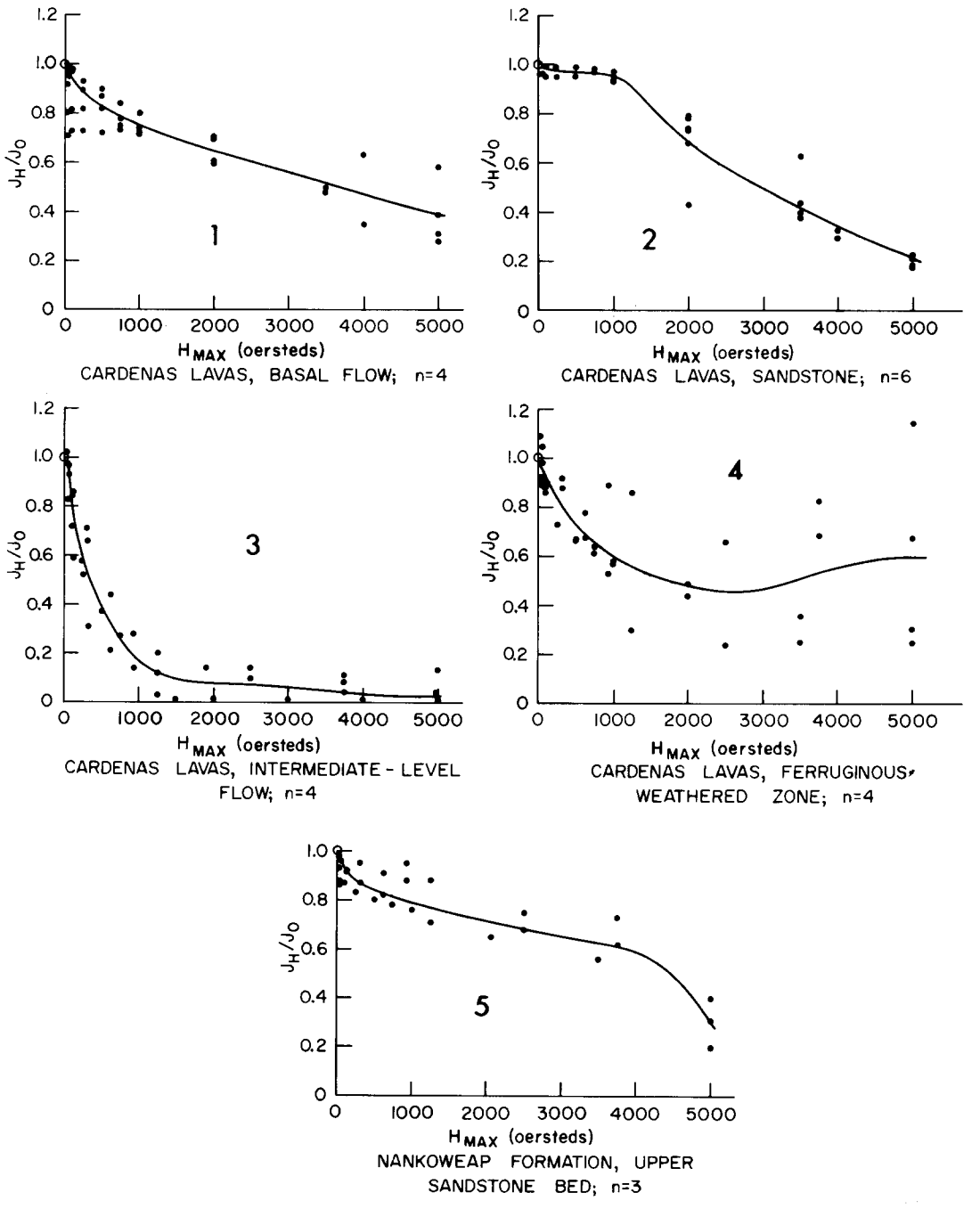


Fig. 4. Normalized alternating field demagnetization curves for five Grand Canyon units sampled. J_H/J_0: ratio of magnetization remaining after a given demagnetization step to original level of magnetization (NRM).

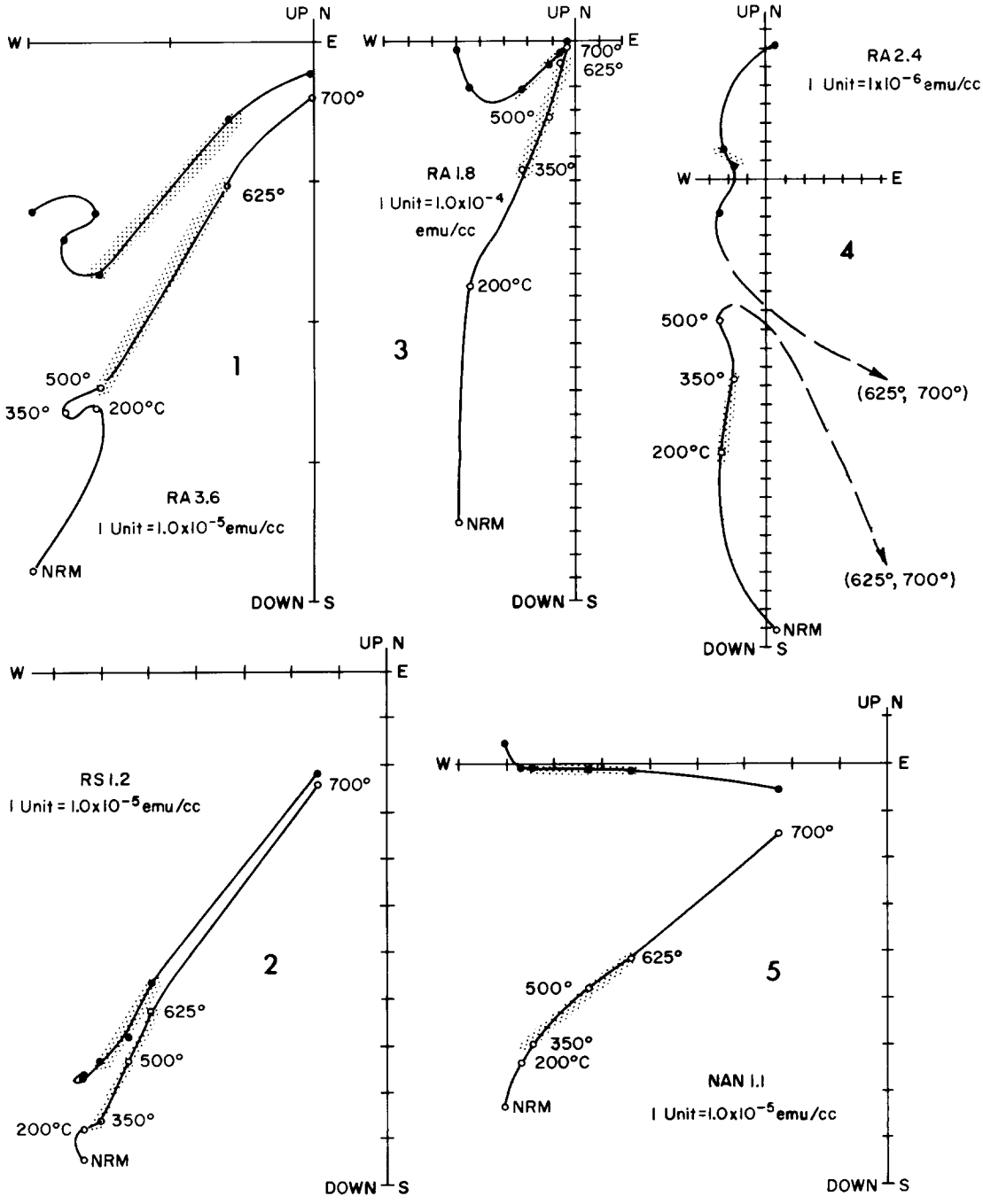


Fig. 5. Representative orthogonal demagnetization diagrams for five Grand Canyon units sampled – progressive thermal demagnetization. Intervals selected for the calculation of poles are stippled. They describe the most stable trends toward the origins, or zero magnetization points, of the diagrams. The plotted points represent successive positions in orthogonal projections of the end of the total magnetization vector during progressive demagnetization (after Zijderveld [16]). Dots denote the projection of the end points of the changing resultant magnetization vector on the horizontal plane; circles, the projections of these end points on the vertical EW plane. Both projections have the EW line in common. The demagnetization temperatures (or intensities of the alternating field, fig. 6) are peak values and are labeled for the vertical projection only. Orientations of specimens are in geographical coordinates.

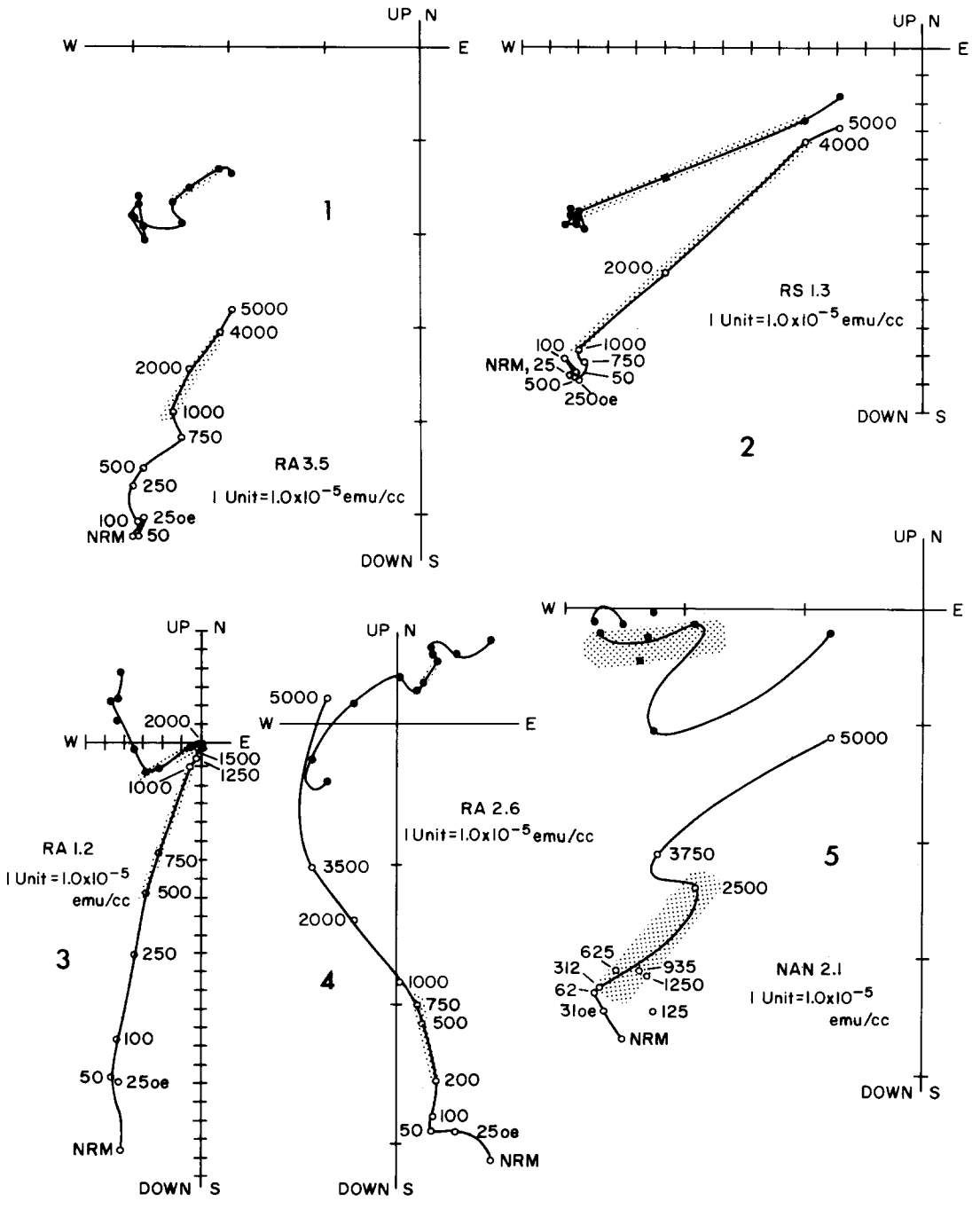


Fig. 6. Representative orthogonal demagnetization diagrams for five Grand Canyon units sampled – progressive alternating-field demagnetization. For explanation of diagrams, see fig. 5.

Table 2
NRM and stable directions of magnetization for samples of the Cardenas Lavas and ferruginous beds of the Nankoweap Formation west of Tanner Canyon, Grand Canyon, Arizona. Structural correction used for all beds: Strike 200°, Dip 15°W. Sampling site, 111°50'W., 36°04'N.

Ident. No.	Stratigraphic Unit		N	Vector Sum R	Declination D	Inclination I	K	α_{95}	Poles Long.	Poles Lat.	δm	δp
5.	Nankoweap Formation, Sandstone Disconformity	NRM	6		276.9	39.0	122.3	6.1	172.6E	18.1N		
		Stable	6	5.953	270.9	36.6	105.4	6.6	174.4E	12.5N	7.7	4.5
4.	Cardenas Lavas Ferruginous weathered zone	NRM	6		323.3	65.8	56.6	9.0	166.2W	60.6N		
		Stable	5	4.949	306.6	65.7	79.4	8.6	167.8W	49.4N	14.1	11.4
3.	Intermediate-level flow	NRM	8		288.1	61.8	206.7	3.9	170.8W	35.8N		
		Stable	7	6.978	258.9	53.6	269.2	3.7	167.8W	11.6N	5.1	3.6
2.	Sandstone	NRM	8		253.5	37.3	54.6	7.6	175.5W	0.3S		
		Stable	8	7.877	253.0	35.9	56.9	7.4	175.9W	1.3S	8.6	5.0
1.	Basal flow	NRM	8		263.7	50.0	154.5	4.5	173.0W	13.0N		
		Stable	8	7.963	253.3	47.9	189.1	4.0	169.1W	4.7N	5.3	3.4
Average, Cardenas Lavas		Stable	10[a]		253.5	38.9	49.8	6.9	174.6W	0.4N	8.2	4.9

a) The eight red sandstone samples of unit 2 are considered to be virtual poles, and the average results for flow units 1 and 3 are considered virtual poles, for $N = 10$.

defined ranges of stability, and minor excursions of the vector are subordinated by the averaging process.

When all stable demagnetization steps for each specimen are used, slightly improved groupings are obtained for the two lava flows and for the weathered zone developed on the intermediate level flow, and slightly increased dispersion is found for the two sandstone units, even though the sandstone is the more stably magnetized. (Details are available on request.) The sandstone samples were collected across narrow stratigraphic intervals, and the slightly increased dispersions, though not statistically significant, might reflect Precambrian secular variations.

AF demagnetization plots of four specimens of the intermediate flow (unit 3, fig. 6) show that three components of magnetization are present in the NRM: (1) a soft component related to the present earth's field; (2) an intermediate component; and (3) a stable component presumably acquired at the time of cooling of the flow. The vector of the intermediate component for four specimens has a declination of 319.5 and an inclination of 50.0 (K = 305, α_{95} = 4). The pole of this component plots at 160.7E, 56.6N after correction for dip and strike of beds, which is distinctly north and west of the NRM and stable poles for the interior flow (table 2; fig. 7). The pole for this intermediate component lies nearest to (about 20 arc degrees west of) the stable pole of the weathered zone, suggesting that the intermediate component of magnetization is related to a pre-Nankoweap weathering which extended at least 20 m below the obvious zone of weathering.

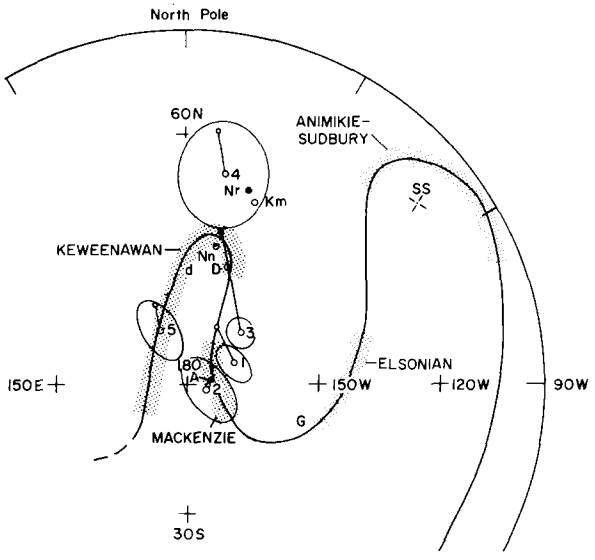


Fig. 7. Stable and NRM poles of five Grand Canyon units sampled, selected Keweenawan poles, and apparent polar wandering trend of Spall. *Unit* (small circles are NRM poles; large circles are stable poles; ovals enclose areas of 95% probability for location of pole). 5. Nankoweap Formation of Maxson [3], two sandstone beds. Cardenas Lavas [10]. 4. Ferruginous weathered zone on intermediate-level flow. 3. Intermediate-level flow. 2. Sandstone above basal flow. 1. Basal flow. A – Average pole for lower part of Cardenas Lavas, upper Unkar Group. G – Average pole for lower part of Unkar Group, Grand Canyon Series, reported by Spall [11]. D – Pole for Duluth Gabbro Complex (Beaver Bay Complex of Schwartz [17], 170.5W, 27.5N [18]). Nn – Mean normal pole, North Shore Volcanic Complex of Gehman [19]; 172.5W, 32.1N [20]. Nr – Mean reversed pole, North Shore Volcanic Complex of Gehman [19]; 161.4W, 45.8N [20]. Km – Mean pole for middle Keweenawan rocks 158.9W, 42.0N [20]. d – Mean pole for 1140–1150 my central Arizona diabase; 179E, 27N [21,22]. ss – Sampling site; 111°50'W, 36°04'N. Apparent polar wandering trend proposed by Spall [11]; named magnetic intervals (stippled bands) are derived from groups of isotopically dated rocks for which paleomagnetic data are available.

5. Discussion

Stable and NRM poles of the five units sampled and an average pole for the Cardenas Lavas (table 2) are shown on fig. 7. Also shown is an apparent polar wandering path proposed by Spall [11] for Precambrian rocks of North America. Poles for flows and a sandstone bed of the Cardenas Lavas plot in and near the Mackenzie magnetic interval (~ 1.2 by; [23]). The pole for the ferruginous weathered zone plots near Palmer's [20] mean pole for middle Keweenawan rocks (Km, fig. 7), northeast of the "early" part of Spall's proposed arc for the Keweenawan magnetic interval (1.2–1.0 by). The Nankoweap pole plots in the lower middle part of the Keweenawan magnetic interval.

Other poles shown on fig. 7 include those for the Duluth Gabbro Complex (D) and normal North Shore Volcanic Complex (Nn) of Gehman [19], which plot in the "very early" part of Spall's proposed arc for the Keweenawan magnetic interval. The mean pole for reversed polarity in the North Shore Complex (Nr),

however, plots distinctly to the north and east, near the pole reported here for the ferruginous weathered zone. The North Shore and Duluth are dated at 1115 ± 15 my by Rb/Sr isochron [24] and 1115 ± 14 my by U/Pb "Concordia" methods [25]. The pole for older (1140–1150 my [26,27]) central Arizona diabase (d) plots in a younger part of Spall's proposed polar wandering path.

A polar wandering path proposed by Robertson and Fahrig [12] differs from Spall's path in an interval that lies between the Mackenzie and Keweenawan magnetic intervals. The discrepancy arises from differences in ages assigned to some rocks whose poles are considered to anchor parts of their proposed paths. The downward leg of the path proposed by Robertson and Fahrig is drawn through Palmer's [20] reversed mean (133.5W, 50.7N) and mean (158.9W, 42.0N) poles for middle Keweenawan rocks, passing near the pole for the weathered zone (unit 4). Neither of the two proposed paths is consistent with the plot for central Arizona diabase. The "early" part of Spall's Keweenawan magnetic interval perhaps should be extended farther to the north and be represented by scattered poles of both normal and reversed polarity.

In spite of the discrepancies, poles from the Grand Canyon Series appear to follow broadly part of the polar wandering path proposed by Spall. If the apparent fit is not an accident, no appreciable rotation has occurred between north-central Arizona and the areas of the North American continent that were used for the definition of Spall's polar wandering path. Spall's path is drawn on the assumption that the various isotopically dated rocks, and areas that serve to define it, were in the same position relative to one another as they are now. An apparent polar wandering path derived from a detailed study of the Grand Canyon Series, anchored by isotopic dating of its flows and sills, should serve as a test of this fundamental assumption.

Acknowledgments

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